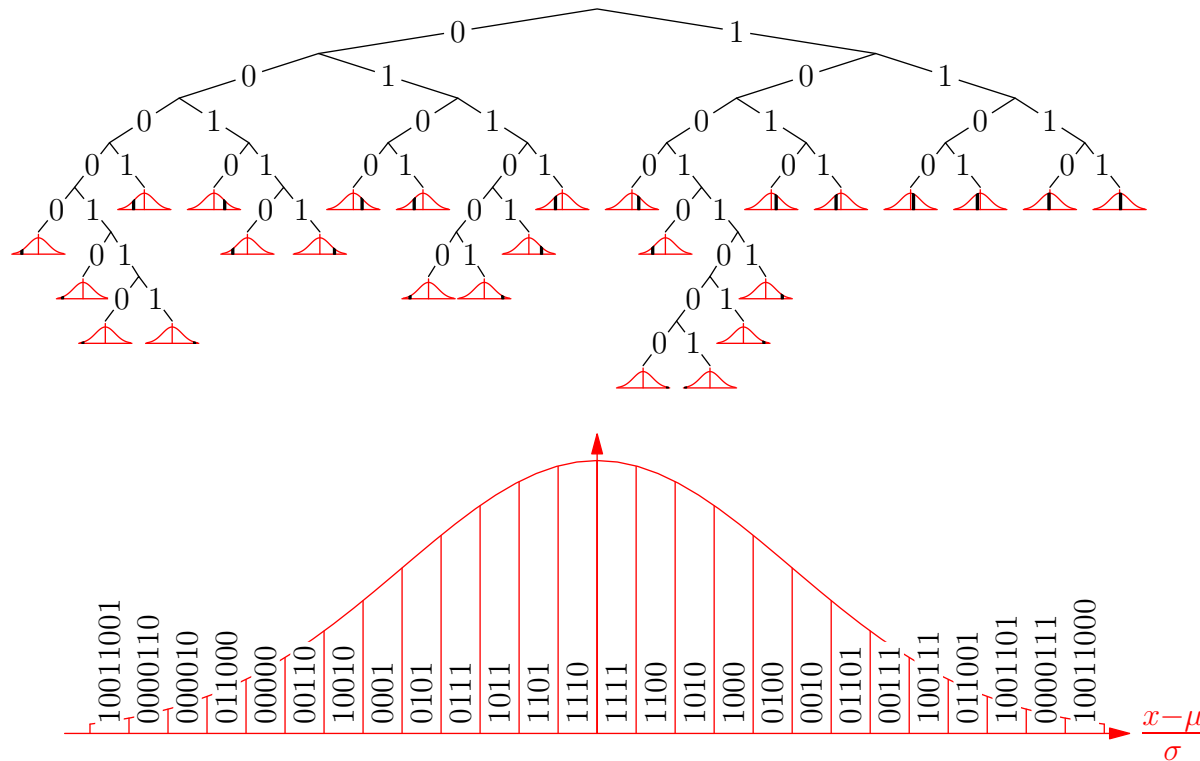


Mathematics of Machine Learning

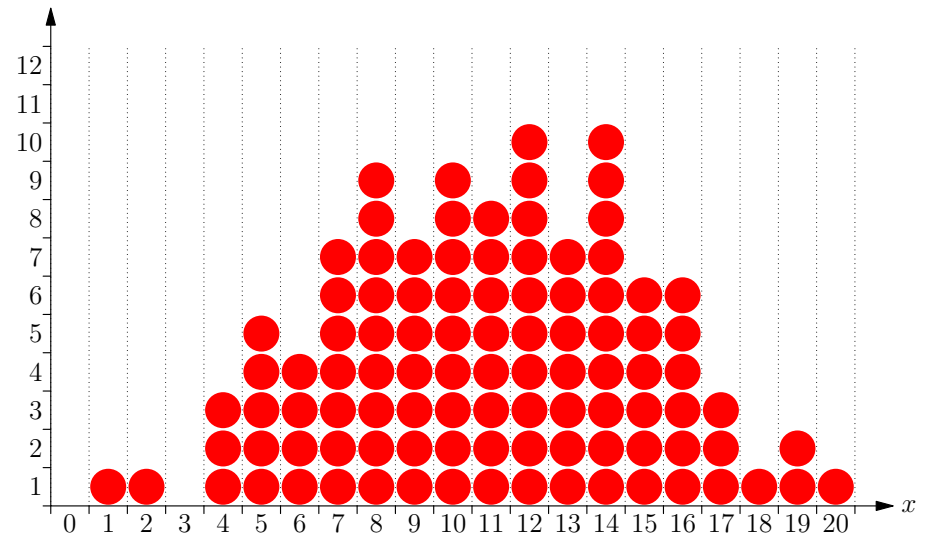
Entropy



Entropy, Coding, Maximum Entropy

Outline

1. **Measuring Uncertainty**
2. Code Length
3. Maximum Entropy



Measuring Uncertainty

- What is more uncertain tossing a coin three times or throwing a dice■
- The answer depends on whether you care about the order of the coin tosses■
- But, how do we answer such a question?■
- Let X be a random variable denoting the possible outcomes■
- Interestingly, Shannon entropy give a precise answer

$$H_X = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x)) \blacksquare$$

Let's Calculate

- For an honest dice $D \in \{1,2,3,4,5,6\}$ and $\mathbb{P}(D = i) = 1/6$ so

$$H_D = -\sum_{i=1}^6 \frac{1}{6} \log_2 \left(\frac{1}{6} \right) = -\log_2 \left(\frac{1}{6} \right) = \log_2(6) \approx 2.584 \text{ bits}$$

- For an honest coin where we care about the order so $C \in \{000,001,\dots,111\}$ the $\mathbb{P}(C = i) = \frac{1}{8}$ and

$$H_C = -\sum_{i=0}^7 \frac{1}{8} \log_2 \left(\frac{1}{8} \right) = -\log_2 \left(\frac{1}{8} \right) = \log_2(8) = 3 \text{ bits}$$

- This clearly makes sense: there are more possible outcomes; all equally likely

Unordered Coin Toss

- What if we don't care about the order of the out-come then $\mathbb{P}(HHH) = \mathbb{P}(TTT) = 1/8$, $\mathbb{P}(HHT) = \mathbb{P}(HTT) = 3/8$ so

$$H_U = -\frac{1}{4}\log_2\left(\frac{1}{8}\right) - \frac{3}{4}\log_2\left(\frac{3}{8}\right) \approx 1.811\text{bits}$$

- This seems reasonable, although it is not obvious how you would determine this without using entropy
- But why Shannon entropy?

Additive Entropy

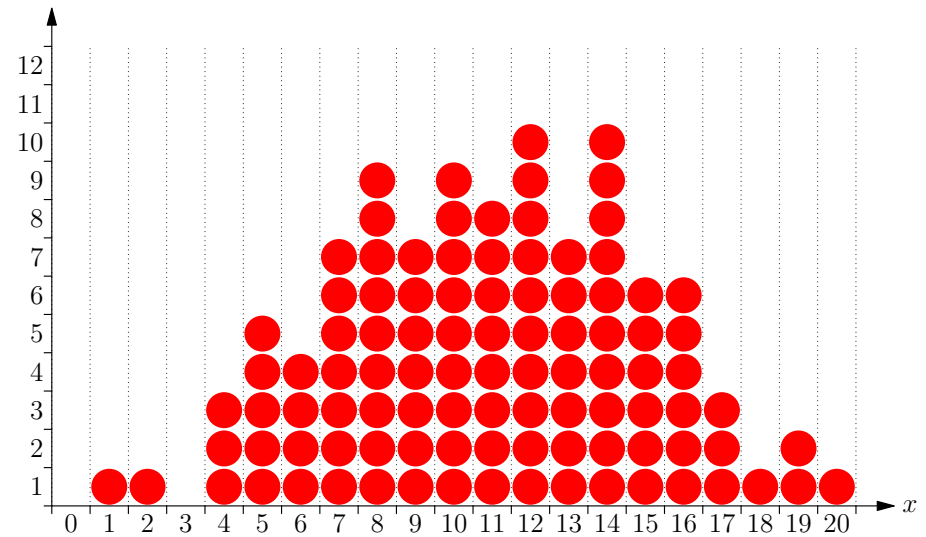
- If H_X and H_Y is the uncertainty of two independent random variable X and Y , what is the uncertainty of the combined event (X,Y) ?

$$\begin{aligned} H_{(X,Y)} &= - \sum_{X,Y} \mathbb{P}(X,Y) \log_2(\mathbb{P}(X,Y)) \blacksquare \\ &= - \sum_{X,Y} \mathbb{P}(X)\mathbb{P}(Y) \log_2(\mathbb{P}(X)\mathbb{P}(Y)) \blacksquare \\ &= - \sum_{X,Y} \mathbb{P}(X)\mathbb{P}(Y) (\log_2(\mathbb{P}(X)) + \log_2(\mathbb{P}(Y))) \blacksquare \\ &= - \sum_X \mathbb{P}(X) \log_2(\mathbb{P}(X)) - \sum_Y \mathbb{P}(Y) \log_2(Y) \blacksquare = H_X + H_Y \blacksquare \end{aligned}$$

- Shannon's entropy is one of the few functions that satisfy this condition $\blacksquare$

Outline

1. Measuring Uncertainty
2. **Code Length**
3. Maximum Entropy



Why Measure Entropy in Bits

- Suppose we had to communicate a message with 2^n equally likely outputs (e.g. the result of n -coin tosses)■
- We can do this with a binary string with n bits (011..0)■
- If there were 5 possible outcomes I could do this with 3 bits, but waste $3/8$ of the message■
- However if we have a batch of 3 independent messages each with 5 outcomes then there are 125 possible outcomes■ We could communicate this with 8 bits. This would waste $3/128$ of the message■
- By batching together enough messages with N outcomes then we asymptotically need just $\log_2(N)$ bits■

Different Probabilities

- We “showed” that if we had N events, X_i , each with probability, $\mathbb{P}(X_i) = 1/N$, we can code the outcomes with a message of length $-\log_2(\mathbb{P}(X_i)) = \log_2(N)$ ■
- With a shorter message we would not be able to distinguish all possible outcomes from the message■
- What happens if some of outcomes occur with a different probability■

X_i :	1	2	3	4	5	6
$p(X_i)$:	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$
Code:	000	001	010	011	10	11
$L = -\log_2(p(X_i))$:	3	3	3	3	2	2

Shannon's Entropy

- If the probabilities are not equal to 2^{-n} we can still find a code with a length very close to $-\log_2(\mathbb{P}(X))$ per message by transmitting a large number of messages■
- The length of the message measures the amount of **surprise** on receiving the message■
- Shannon's entropy is the expected length of the message to communicate a random variable X

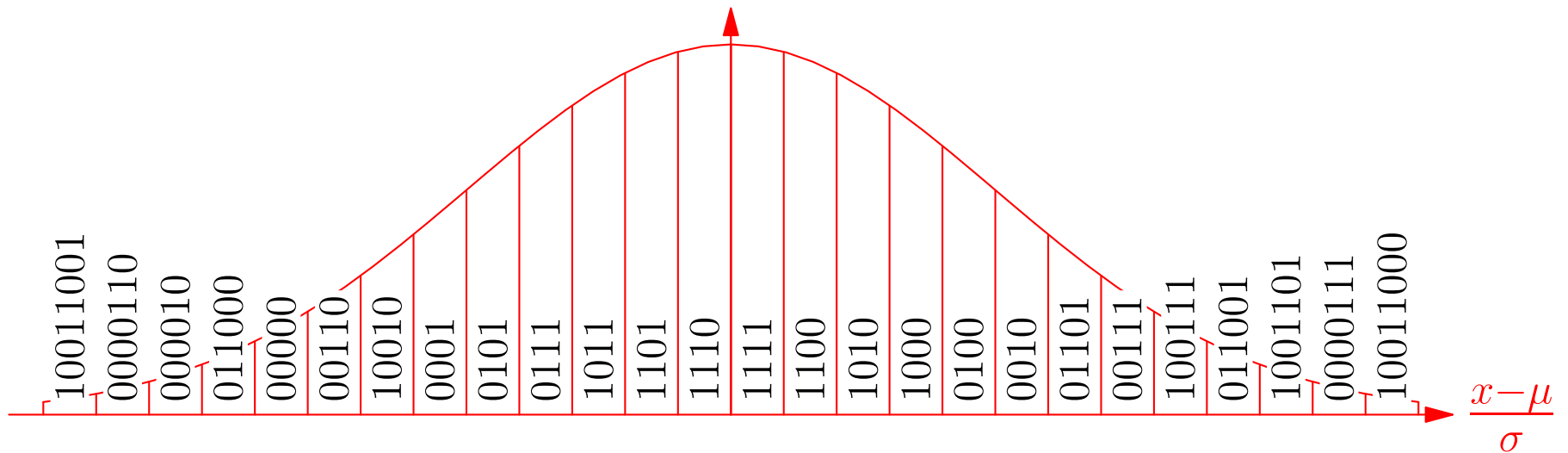
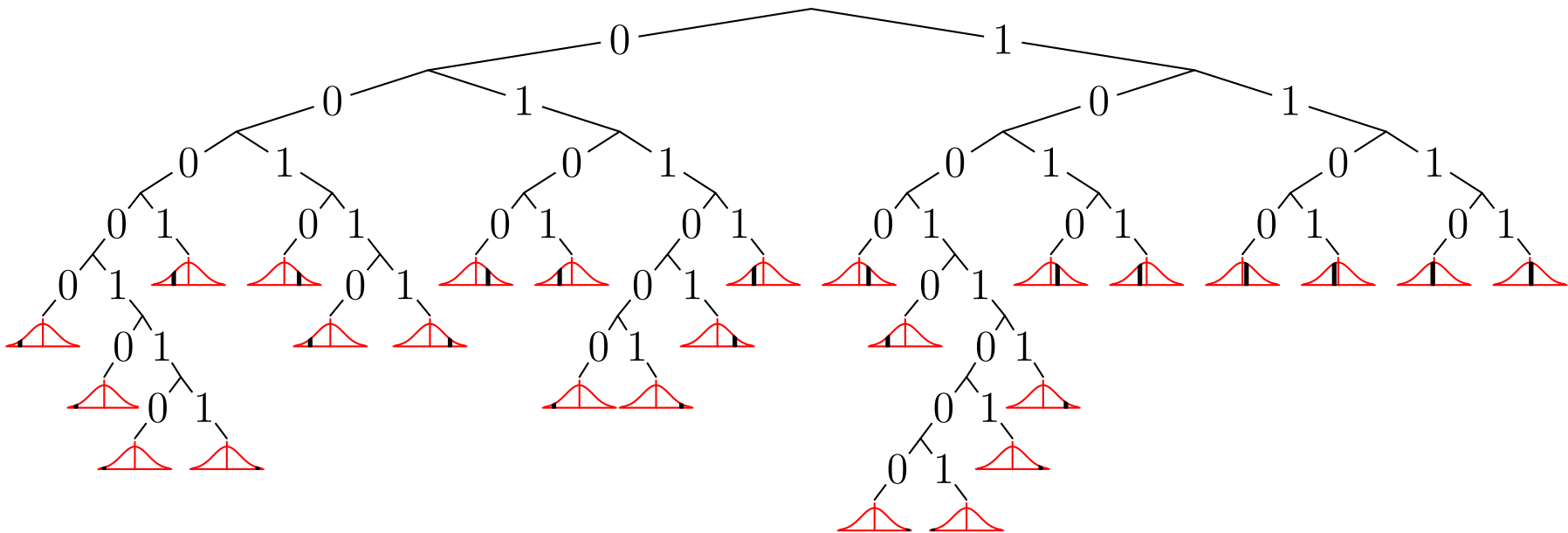
$$H_X = \mathbb{E}_X[-\log_2(\mathbb{P}(X))] = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x))■$$

- The expected length is a measure of the uncertainty■(how much information on average we need to convey the outcome)■

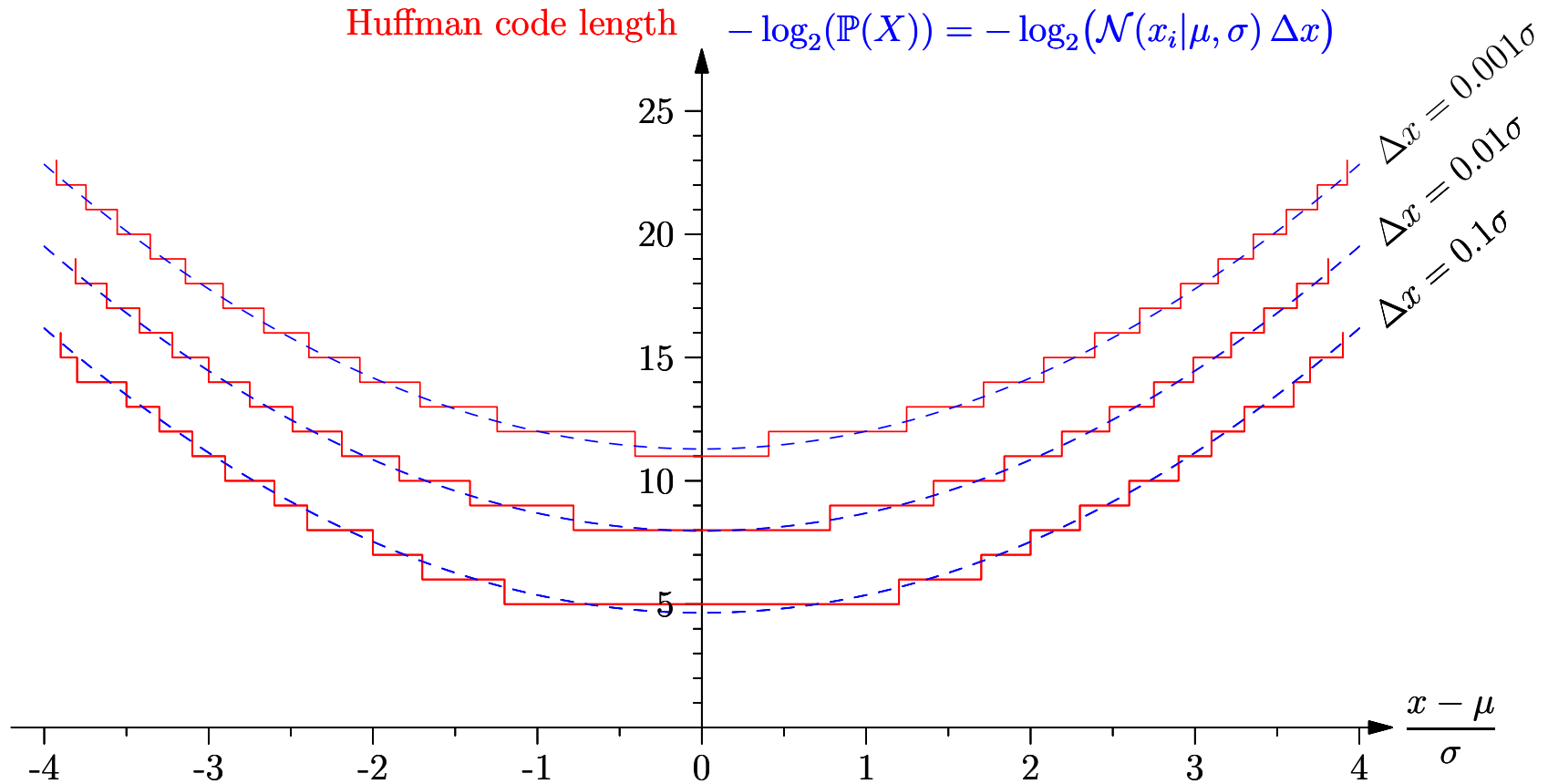
Real Codes

- Those of you with some computer science background will realise that we can't actually use different length strings in a code without paying some price■
- We won't know where a code word ends so we can't decode the message■
- An optimal solution is to use Huffman encoding where we associate the leaf of a tree with each code word■
- Using the tree we can decode any message constructed using the tree■
- There is a greedy algorithm for constructing the optimal tree■

Coding Normals



Coding Normals to Accuracy Δx



bits and nats

- We have measured entropy in **bits** using

$$H_X = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x)) \blacksquare$$

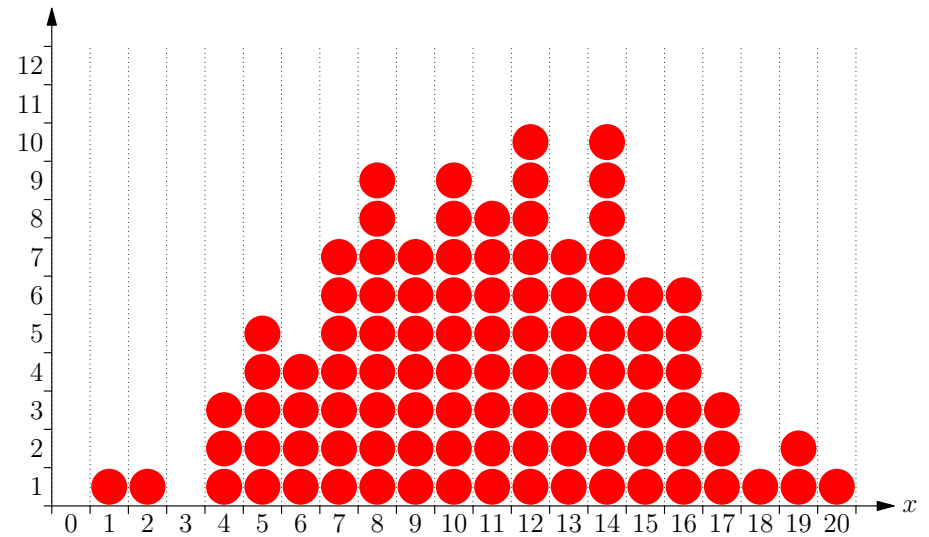
- Sometimes it is easier to use natural logarithms

$$H_X = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \ln(\mathbb{P}(X = x)) \blacksquare$$

- In this case the entropy is measured in **nats** with 1 nat equal to $\log_2(e)$ bits $\blacksquare$
- This is often easier when we want to do calculus on entropy $\blacksquare$

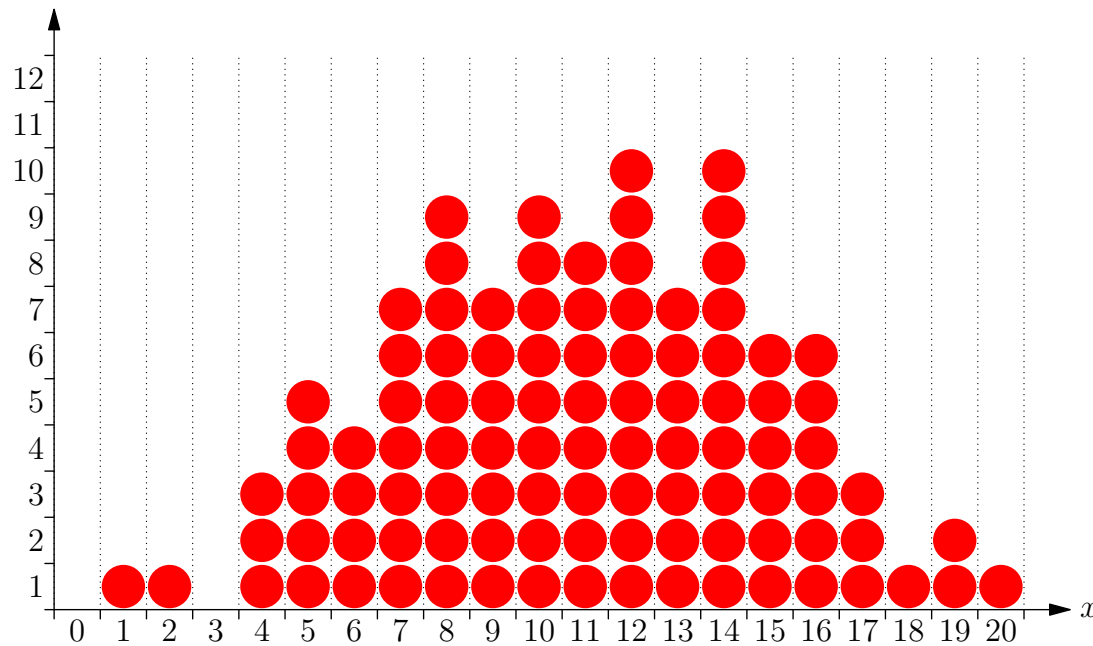
Outline

1. Measuring Uncertainty
2. Code Length
3. **Maximum Entropy**



Number of States

- Suppose I have N balls I then put in K boxes so that n_i balls finish up in box i



- Counting the number of ways to do this

$$\mathbb{P}(\mathbf{n}) = \frac{N!}{n_1!n_2!\cdots n_K!}$$

Stirling's Approximation

- We can approximate the factorial $n!$ using **Stirling's approximation**

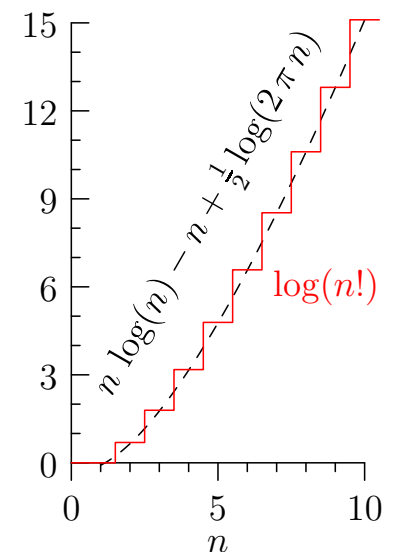
$$n! \approx \sqrt{2\pi n} n^n e^{-n}$$

$$\log(n!) = n \log(n) - n + \frac{1}{2} \log(2\pi n)$$

- Using this in our formula for $\mathbb{P}(\mathbf{n})$ we have

$$\begin{aligned} \mathbb{P}(\mathbf{n}) &= \frac{N!}{n_1! n_2! \cdots n_K!} = e^{\log(N!) - \sum_i \log(n_i!)} \\ &\approx C e^{N \log(N) - \sum_i n_i \log(n_i)} \approx C e^{-N \sum_i \frac{n_i}{N} \log\left(\frac{n_i}{N}\right)} \end{aligned}$$

Order n terms cancel as $N = \sum_i n_i$



Number of States and Entropy

- Let $p(x_i) = n_i/N$ be the proportion of balls in bin i then

$$\mathbb{P}(\mathbf{n}) \approx C e^{N H_X}$$

where

$$H_X = - \sum_i p(x_i) \log(p(x_i)) \blacksquare$$

- That is, the “entropy” can be seen as a measure of the logarithm of the number of configurations $\blacksquare$
- When the number of balls, $N \rightarrow \infty$ the overwhelmingly likely configurations is the one that maximises the entropy subject to the observed mean and variance $\blacksquare$

Maximum Entropy Method

- When we are trying to infer a distribution given some observations then we can maximise the entropy subject to constraints—the entropy acts as a prior
- This is known as the **maximum entropy method**
- We can rationalise this as this is by far the most likely set of configurations consistent with the observations
- Alternatively we can see this as maximising our uncertainty given what we know—being as unbiased as possible
- It only gives a good approximation if all possibilities are equally likely

Knowing the Mean and Variance

- Consider a continuous random variable, X , with a known mean and second moment

$$\mathbb{E}[X] = \mu, \quad \mathbb{E}[X^2] = \mu_2 = \mu^2 + \sigma^2$$

- To maximise the entropy subject to constraints consider

$$\begin{aligned} \mathcal{L}(f) = & - \int f_X(x) \log(f_X(x)) dx + \lambda_0 \left(\int f_X(x) dx - 1 \right) \\ & + \lambda_1 \left(\int f_X(x) x dx - \mu \right) + \lambda_2 \left(\int f_X(x) x^2 dx - \mu_2 \right) \end{aligned}$$

- Thus

$$\frac{\delta \mathcal{L}(f)}{\delta f_X(x)} = -\log(f_X(x)) - 1 + \lambda_0 + \lambda_1 x + \lambda_2 x^2 = 0$$

- Or

$$f_X(x) = e^{-1+\lambda_0+\lambda_1 x+\lambda_2 x^2}$$

Normal Distribution

- We have three constraints

$$\begin{aligned}\int e^{-1+\lambda_0+\lambda_1 x+\lambda_2 x^2} dx &= 1 \\ \int e^{-1+\lambda_0+\lambda_1 x+\lambda_2 x^2} x dx &= \mu \\ \int e^{-1+\lambda_0+\lambda_1 x+\lambda_2 x^2} x^2 dx &= \mu_2 = \mu^2 + \sigma^2\end{aligned}$$

- Solving for λ_0 , λ_1 and λ_2 then

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$$

- That is, the normal distribution is the maximum entropy distribution given we known the mean and variance

Using Maximum Entropy

- Maximum entropy is often used to infer distributions■
- It can be very effective, but it might not work well if there are other constraints that we have not included■
- The place that they work superbly well is in statistical physics■
- The whole of statistical physics is about inferring distributions making observations of volume, pressure, etc.■
- Temperature appears rather strangely as a Lagrange multiplier■

Historic Entropy

- Historically entropy was first introduced in statistical physics by Rudolf Clausius in 1865 (although Macquorn Rankine discussed it in 1850)■
- Its interpretation as the number of states was introduced by Ludwig Boltzmann■
- The person who got it all right was Josiah Willard Gibbs (and James Clerk Maxwell)■
- Claude Shannon invented information theory base on entropy around 1948 (more on that in the next lecture)■
- Ed Jaynes was the first to understand that statistical physics can be seen as an inference problem■

Conclusion

- Entropy provides a measure of the disorder or uncertainty in a system■
- It forms the basis of information theory which we will look at in the next lecture■
- $-\log(\mathbb{P}(X = x))$ can be seen as the minimum length of a message to communicate x ■
- This will be used as the basis of the minimum description length formalism also discussed in the next lecture■
- Entropy can be used as a prior, which we often maximise subject to constraints to obtain an unbiased estimate■